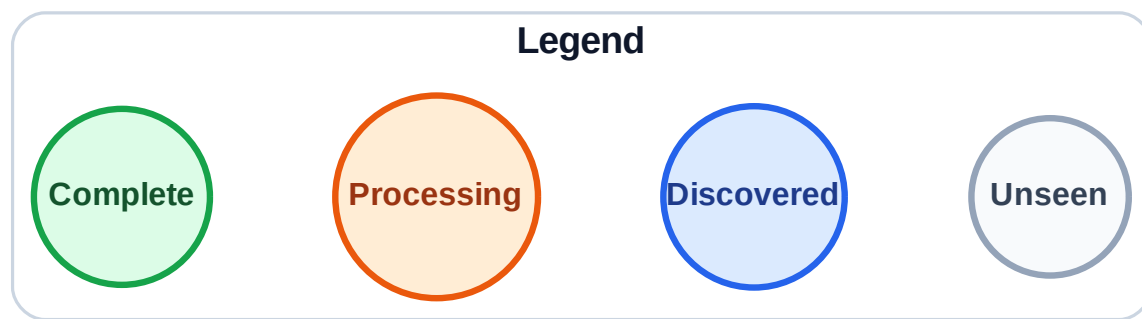


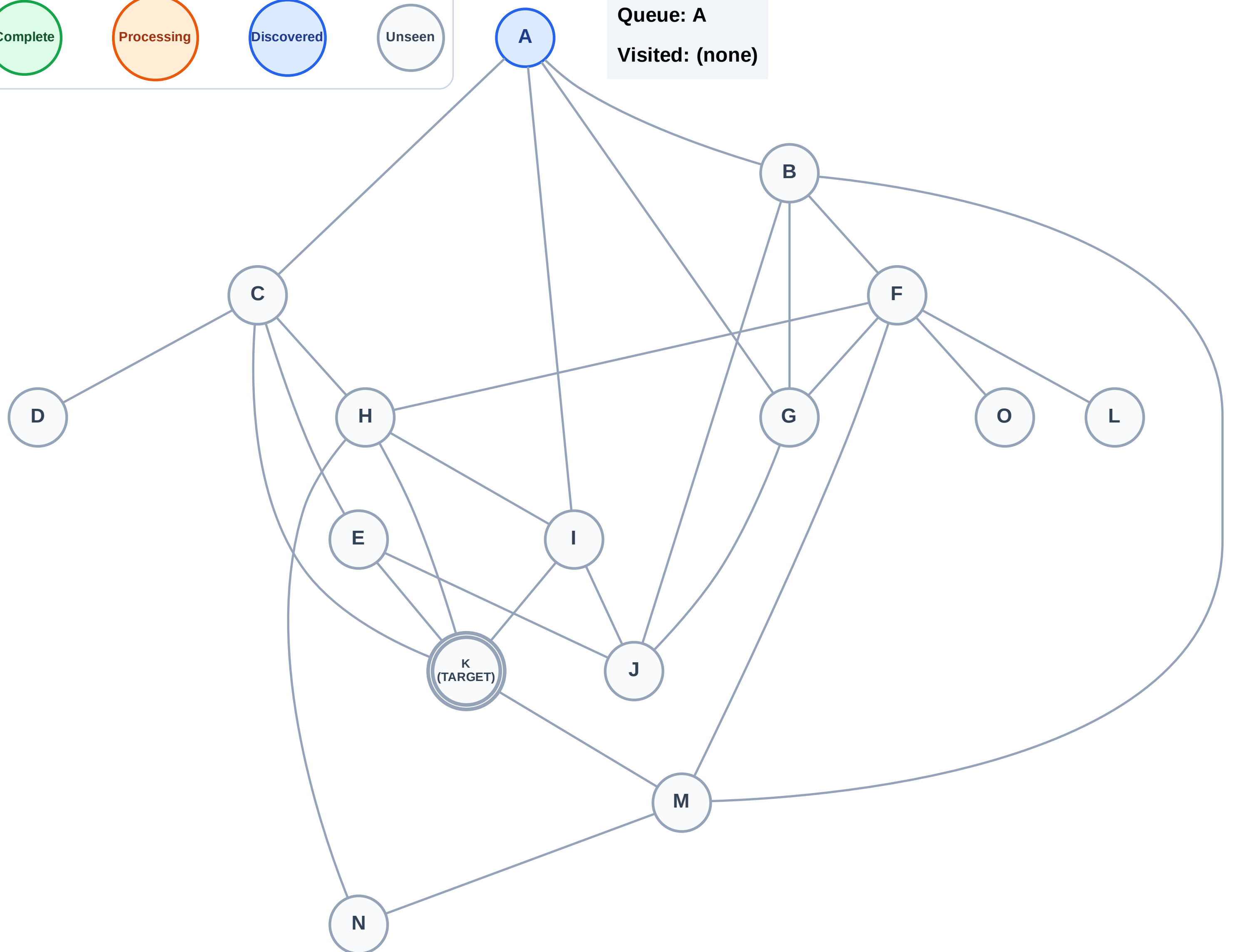
BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))



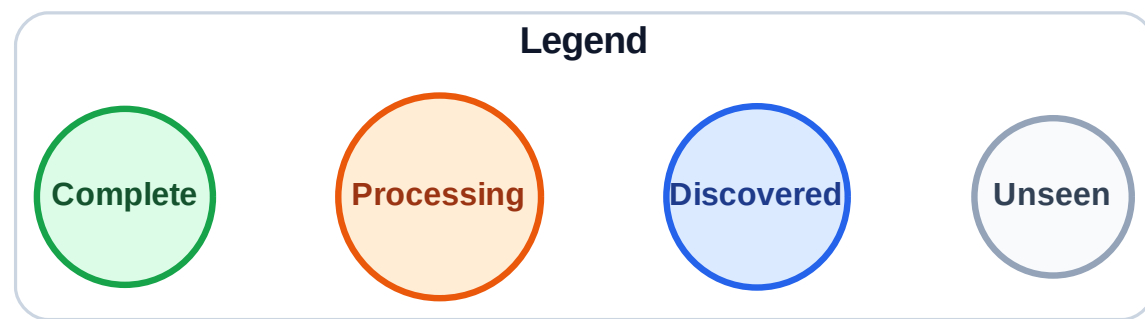
Queue: A

Visited: (none)



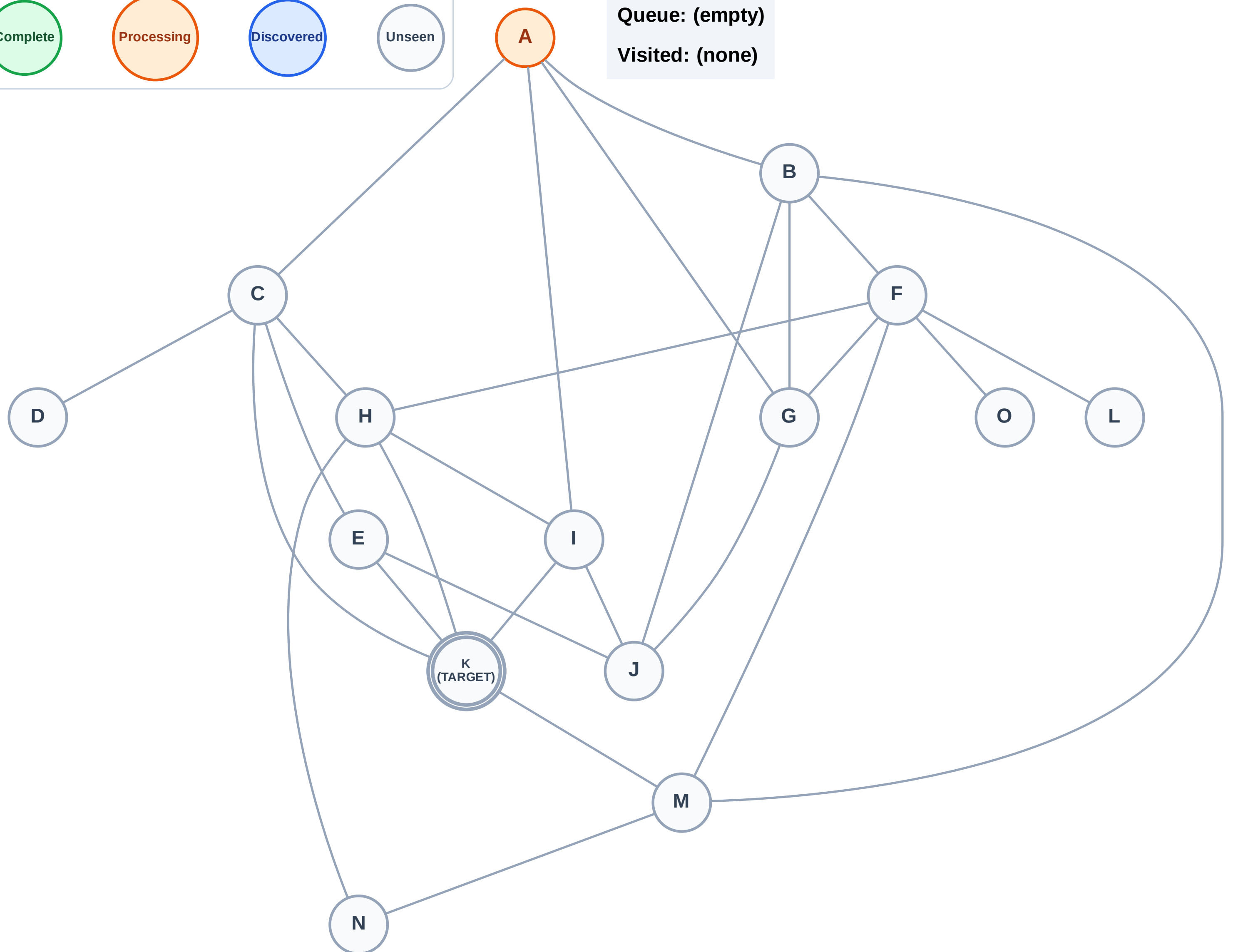
BFS Traversal

Step 2: Process node A



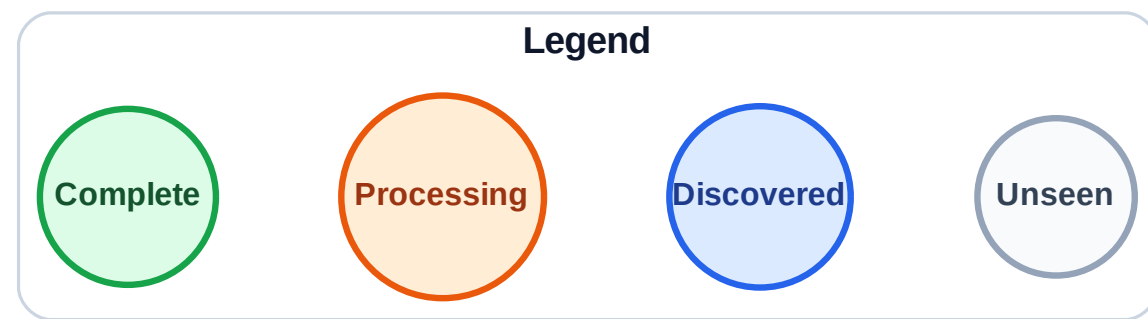
Queue: (empty)

Visited: (none)



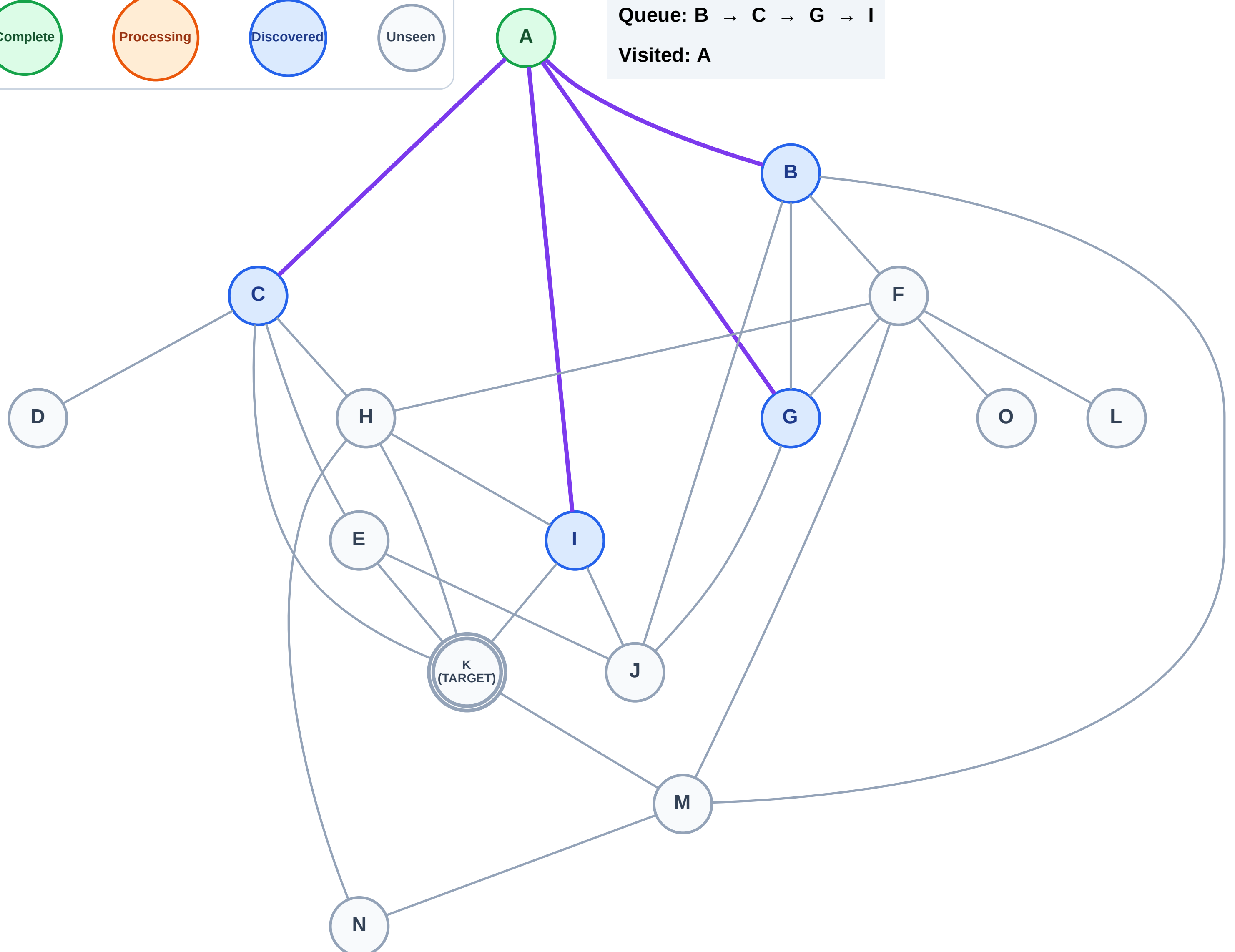
BFS Traversal

Step 3: Discover B, C, G, I from A



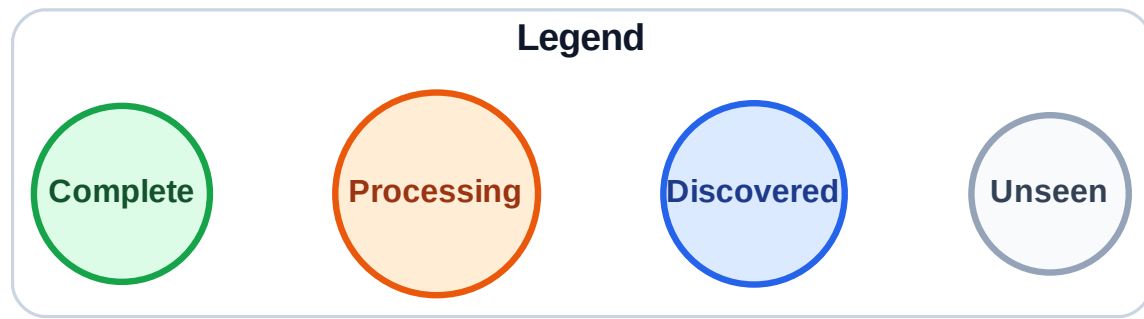
Queue: B → C → G → I

Visited: A



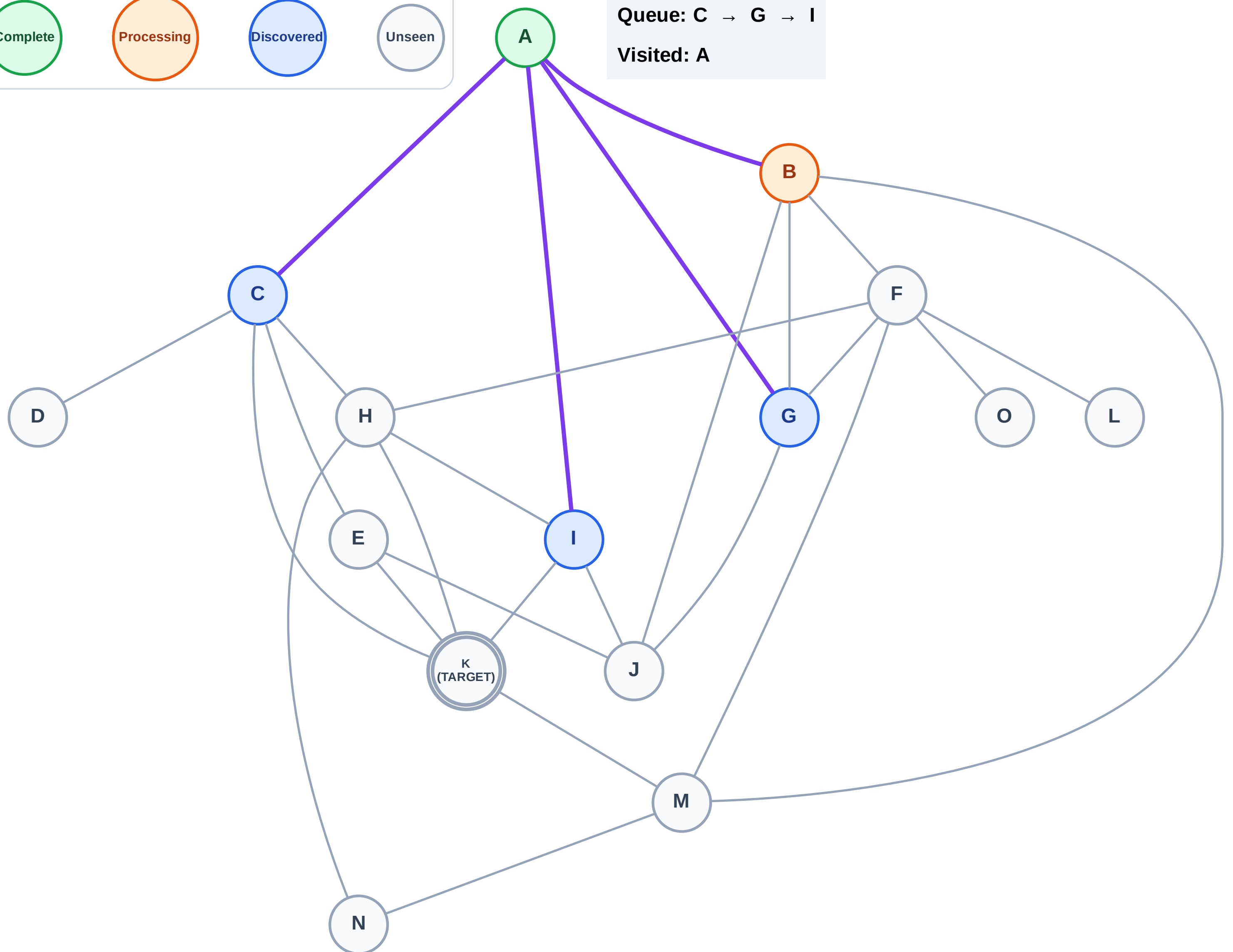
BFS Traversal

Step 4: Process node B



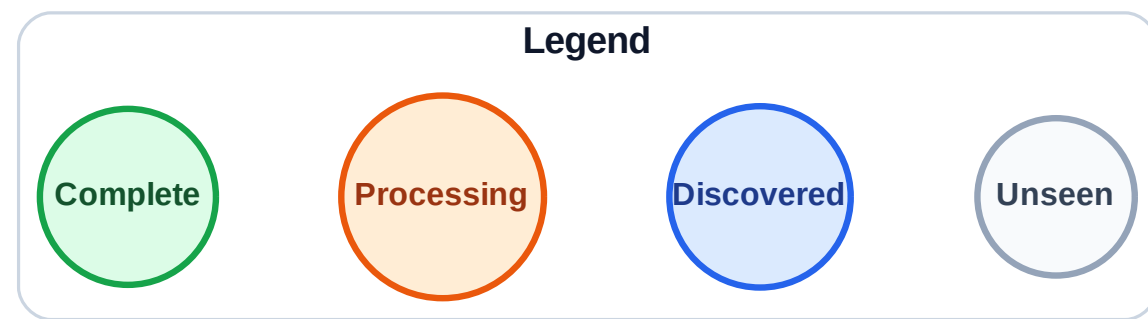
Queue: C → G → I

Visited: A



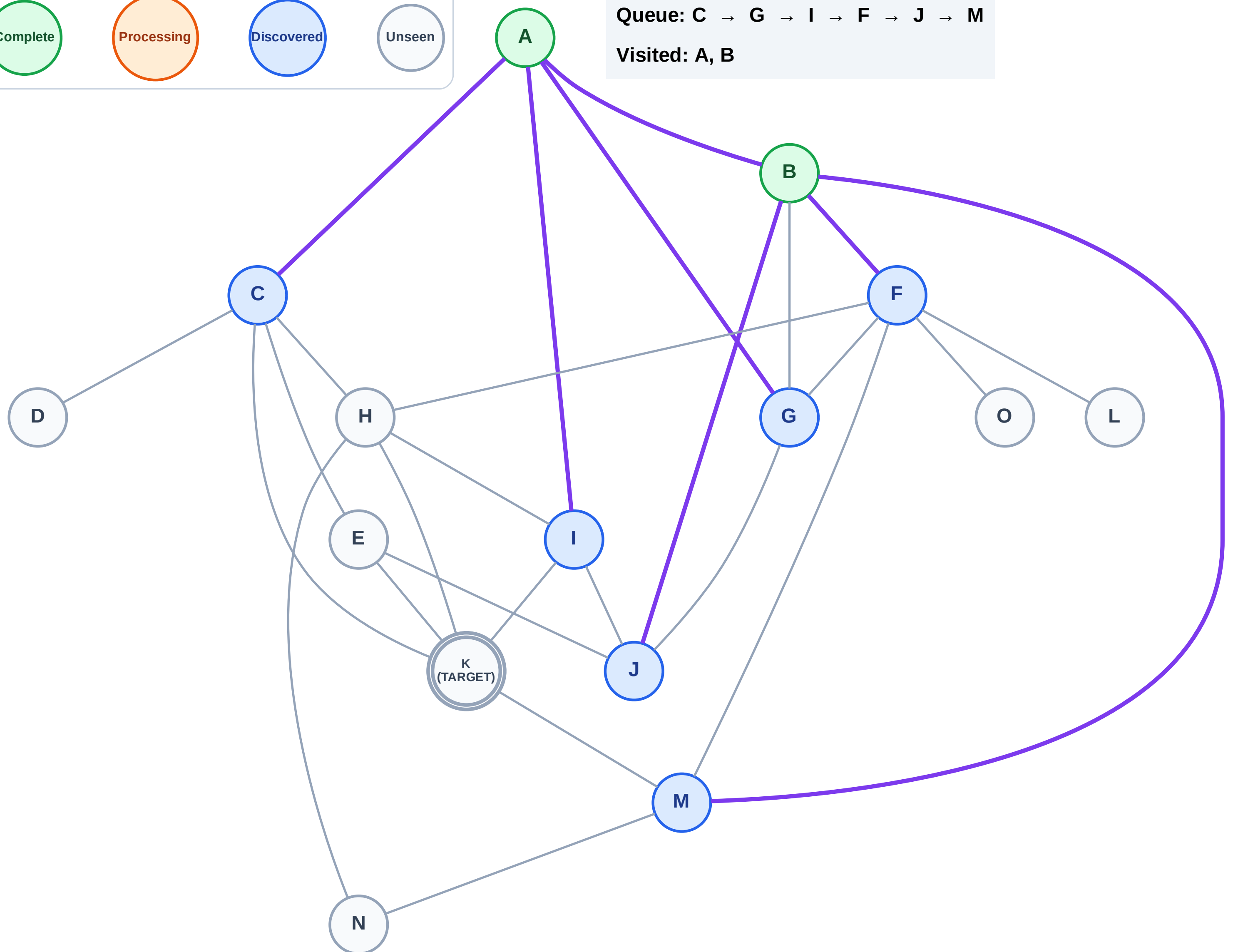
BFS Traversal

Step 5: Discover F, J, M from B



Queue: C → G → I → F → J → M

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

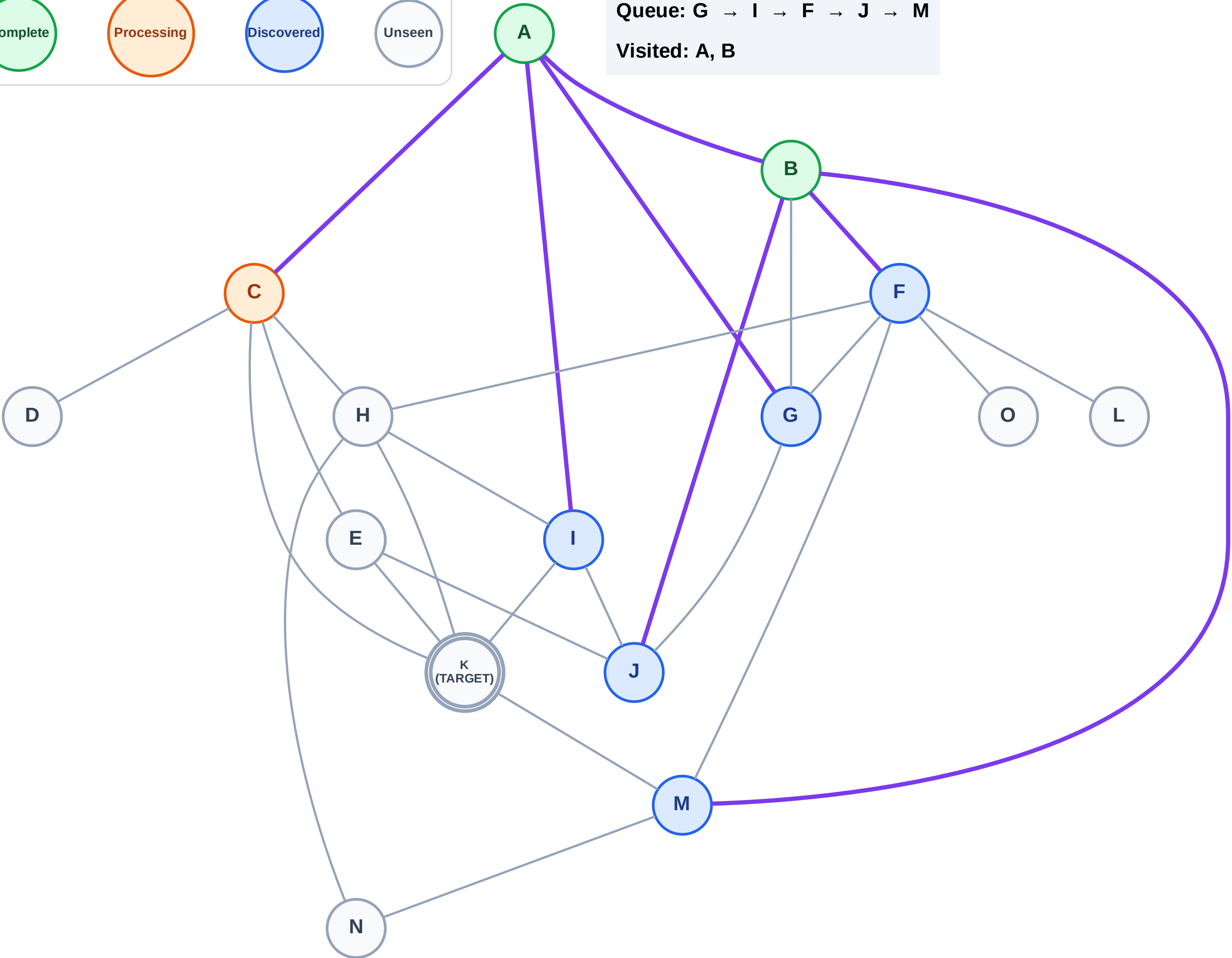
Processing

Discovered

Unseen

Queue: G → I → F → J → M

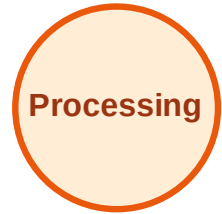
Visited: A, B



BFS Traversal

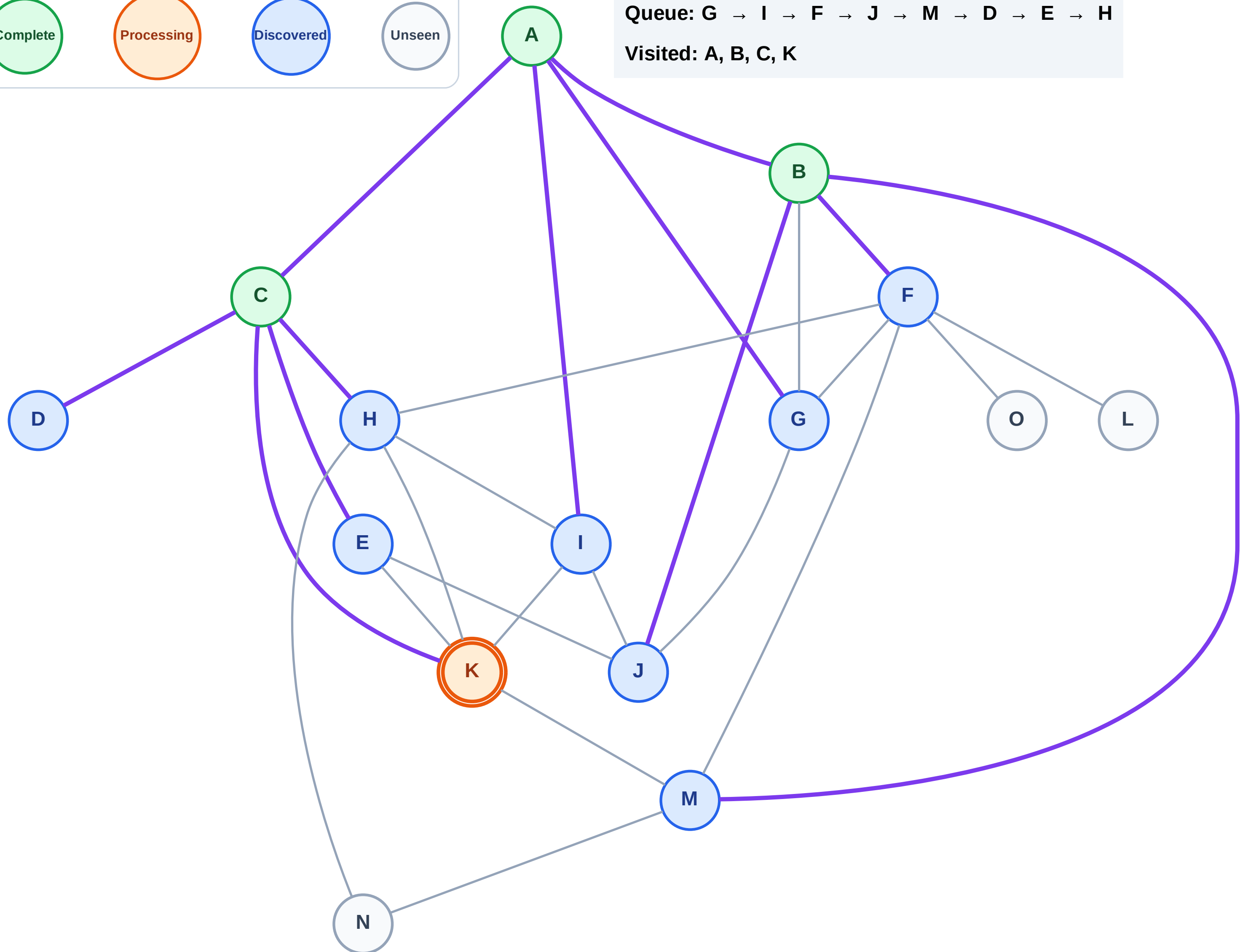
Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → C → K

Legend



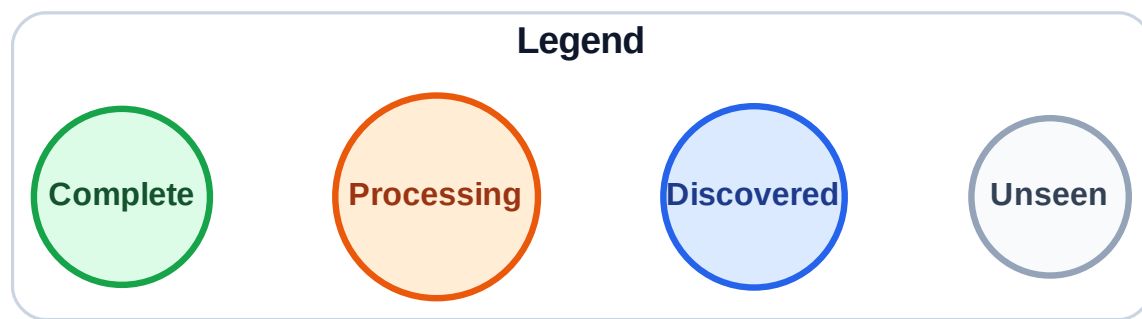
Queue: G → I → F → J → M → D → E → H

Visited: A, B, C, K

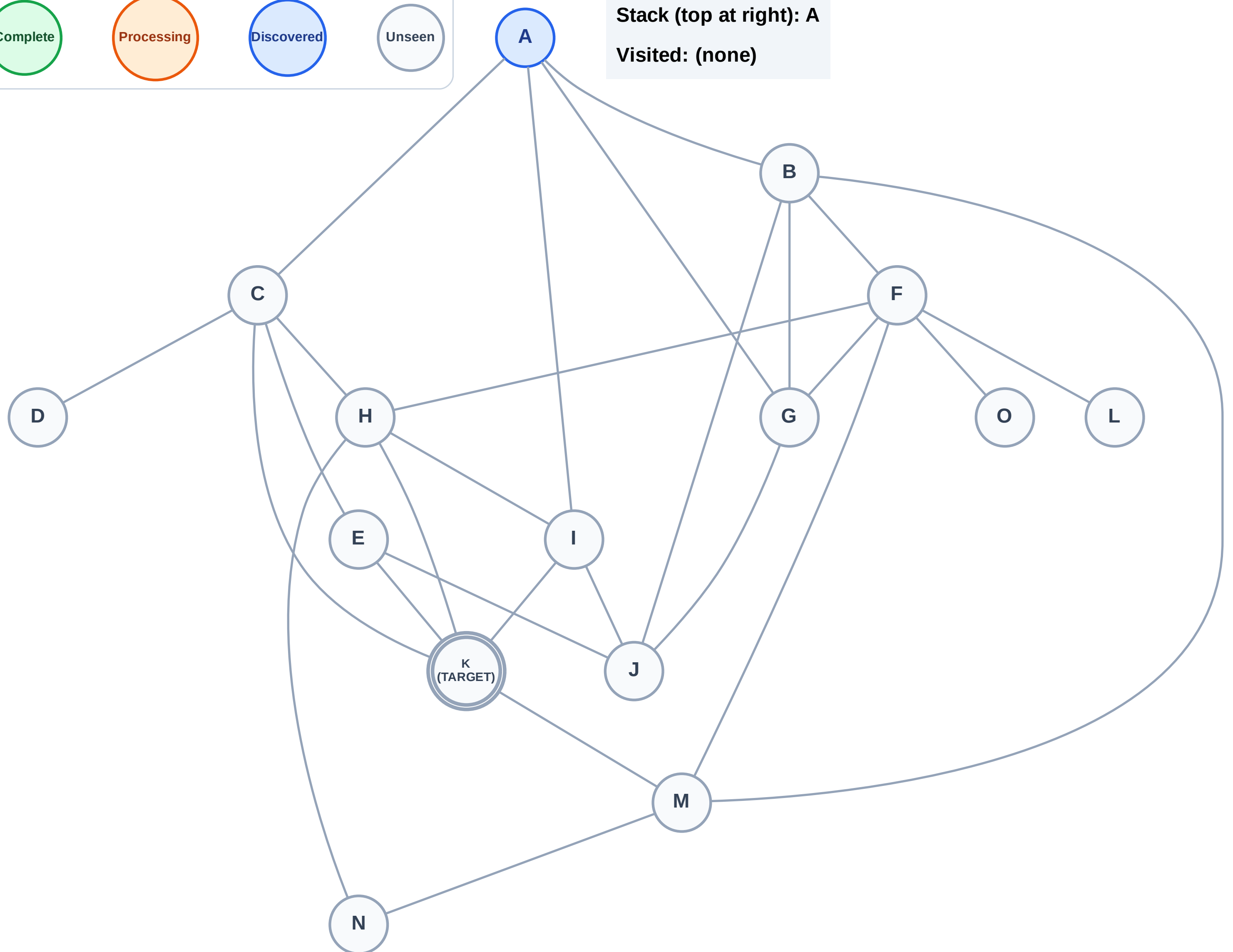


DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))



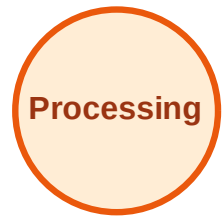
Stack (top at right): A
Visited: (none)



DFS Traversal

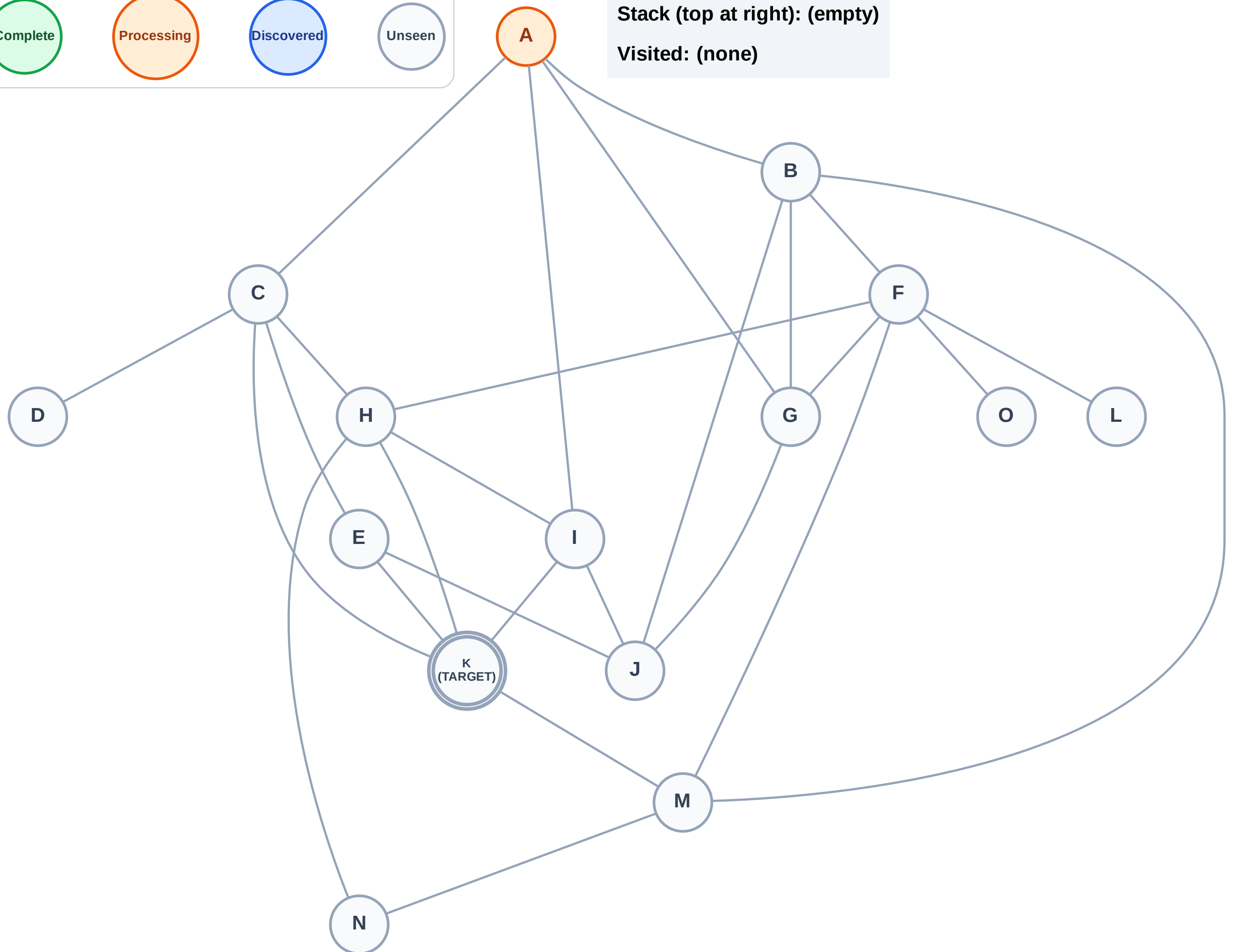
Step 2: Process node A

Legend



Stack (top at right): (empty)

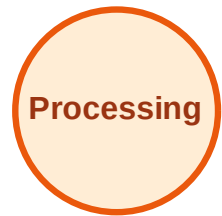
Visited: (none)



DFS Traversal

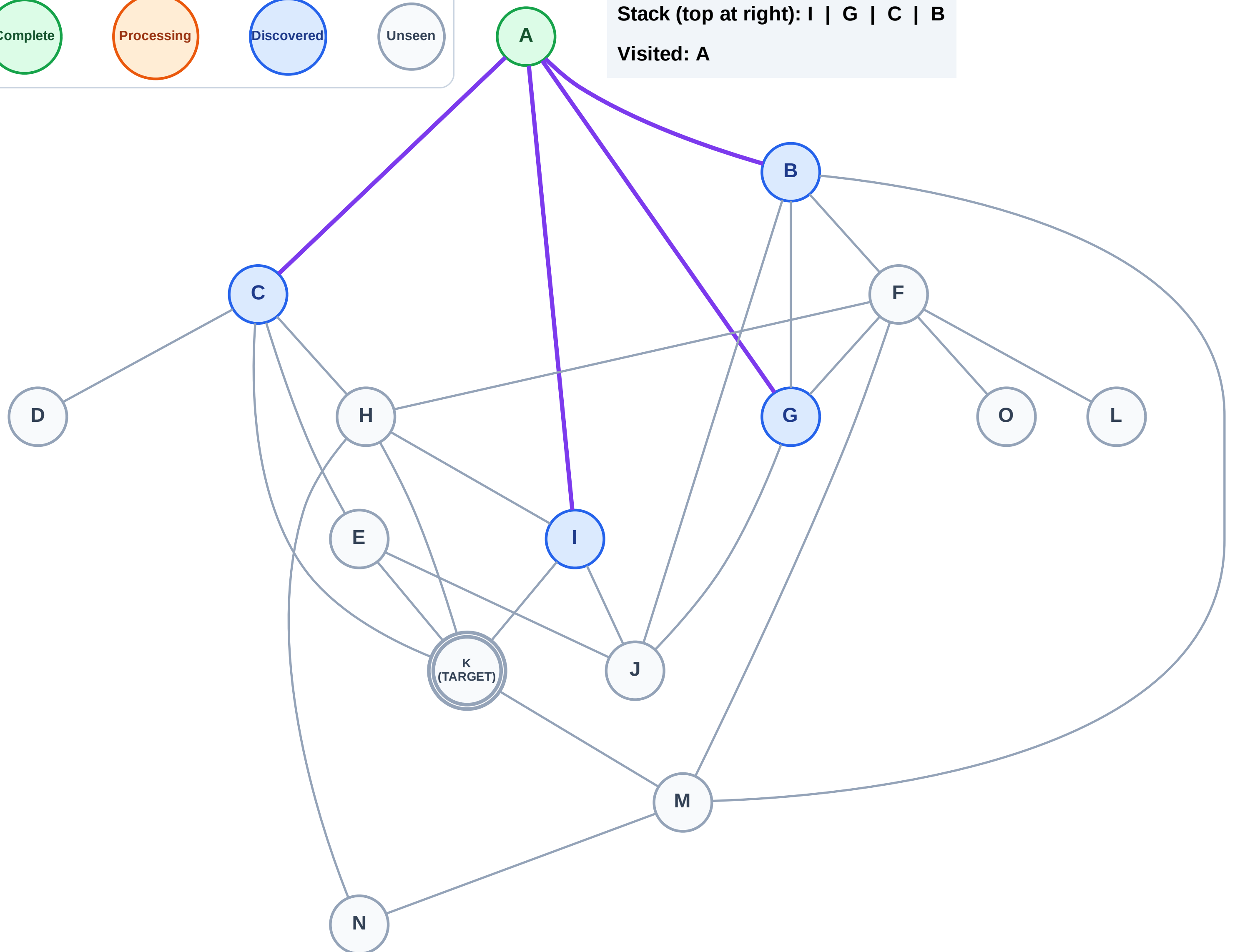
Step 3: Discover I, G, C, B from A

Legend



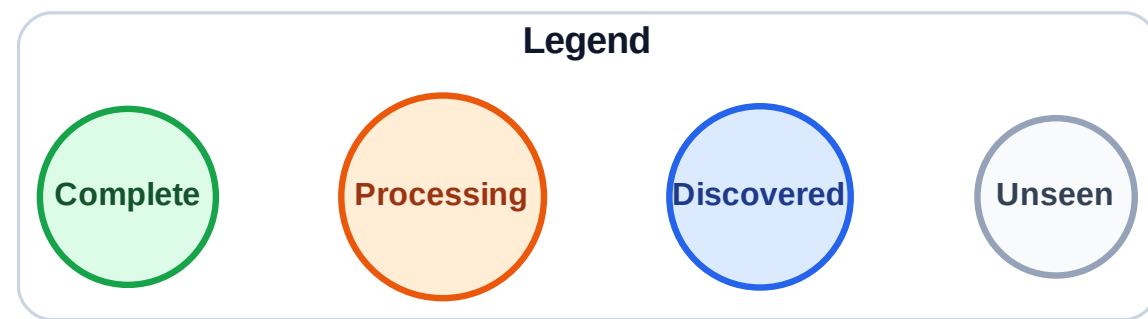
Stack (top at right): I | G | C | B

Visited: A



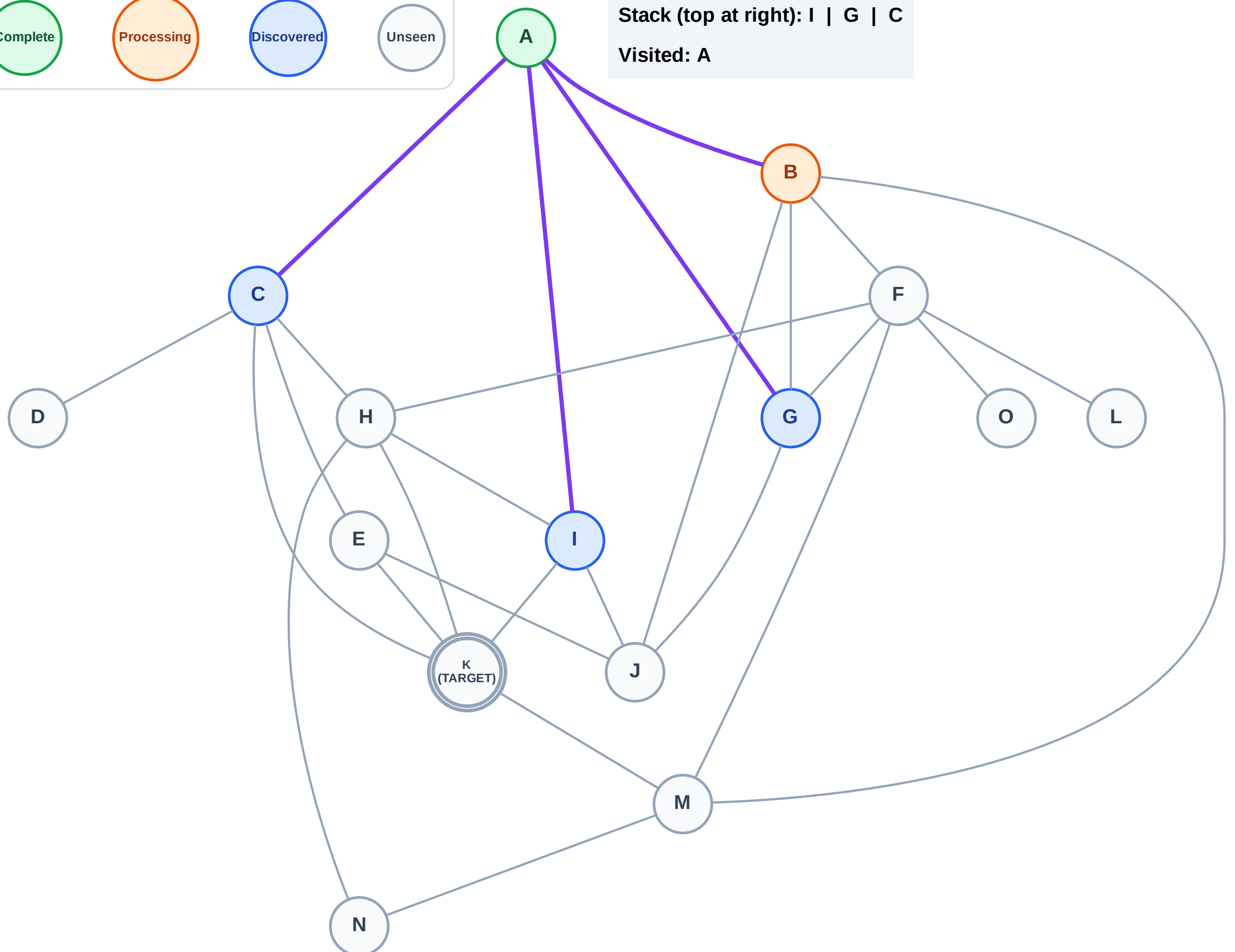
DFS Traversal

Step 4: Process node B



Stack (top at right): I | G | C

Visited: A



DFS Traversal

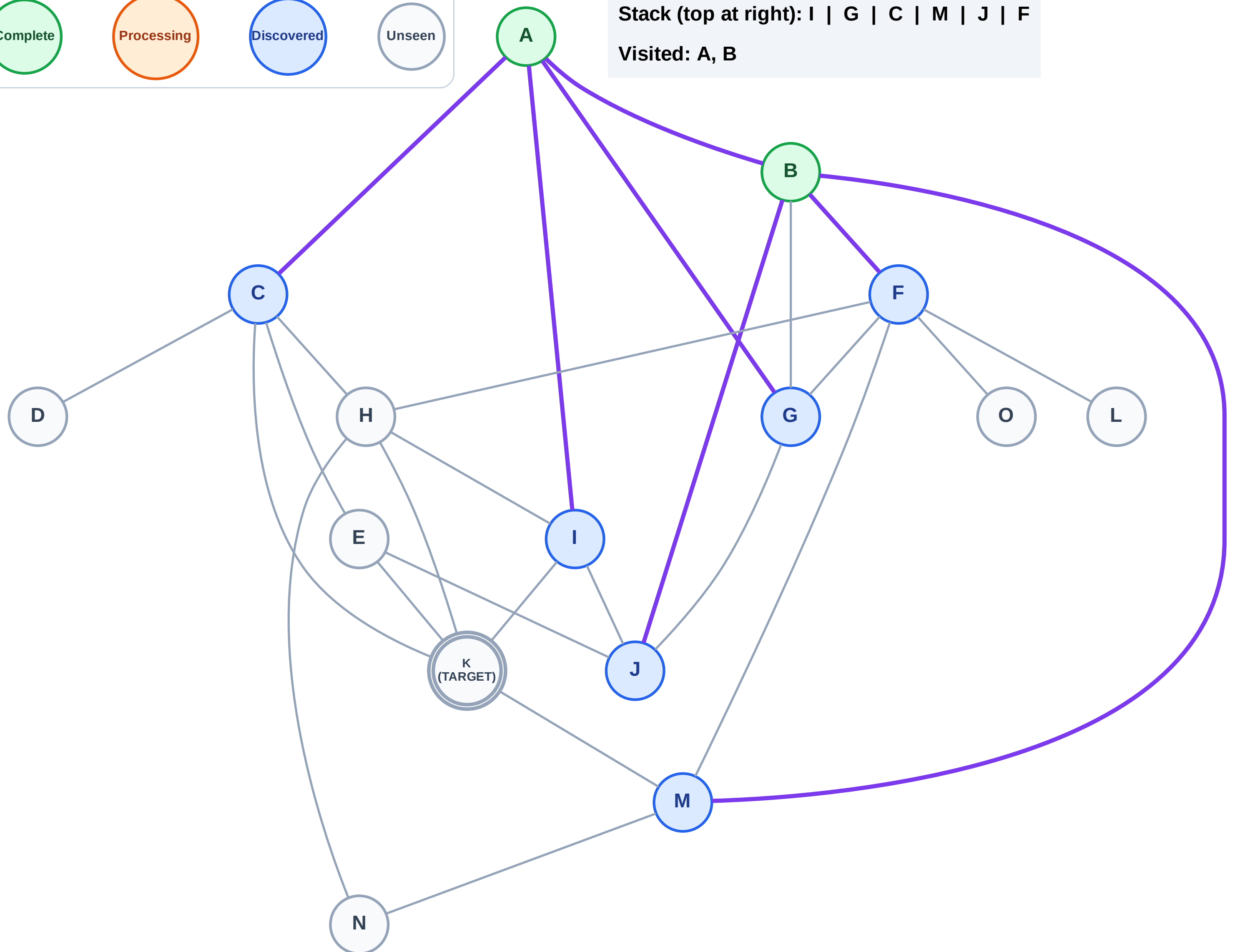
Step 5: Discover M, J, F from B

Legend



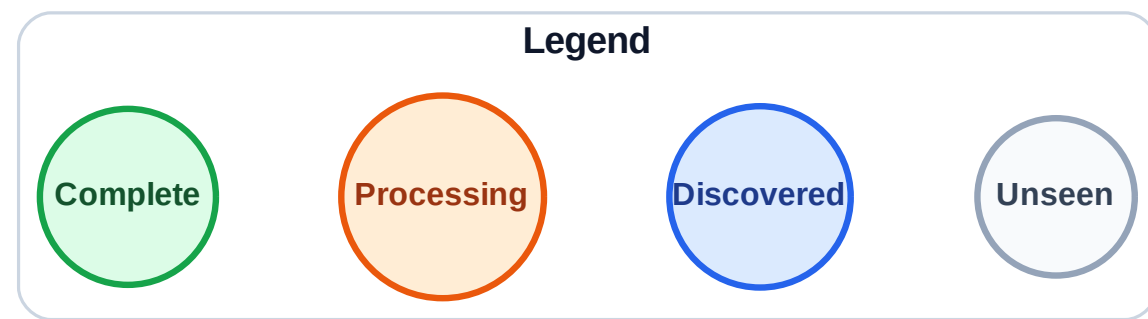
Stack (top at right): I | G | C | M | J | F

Visited: A, B



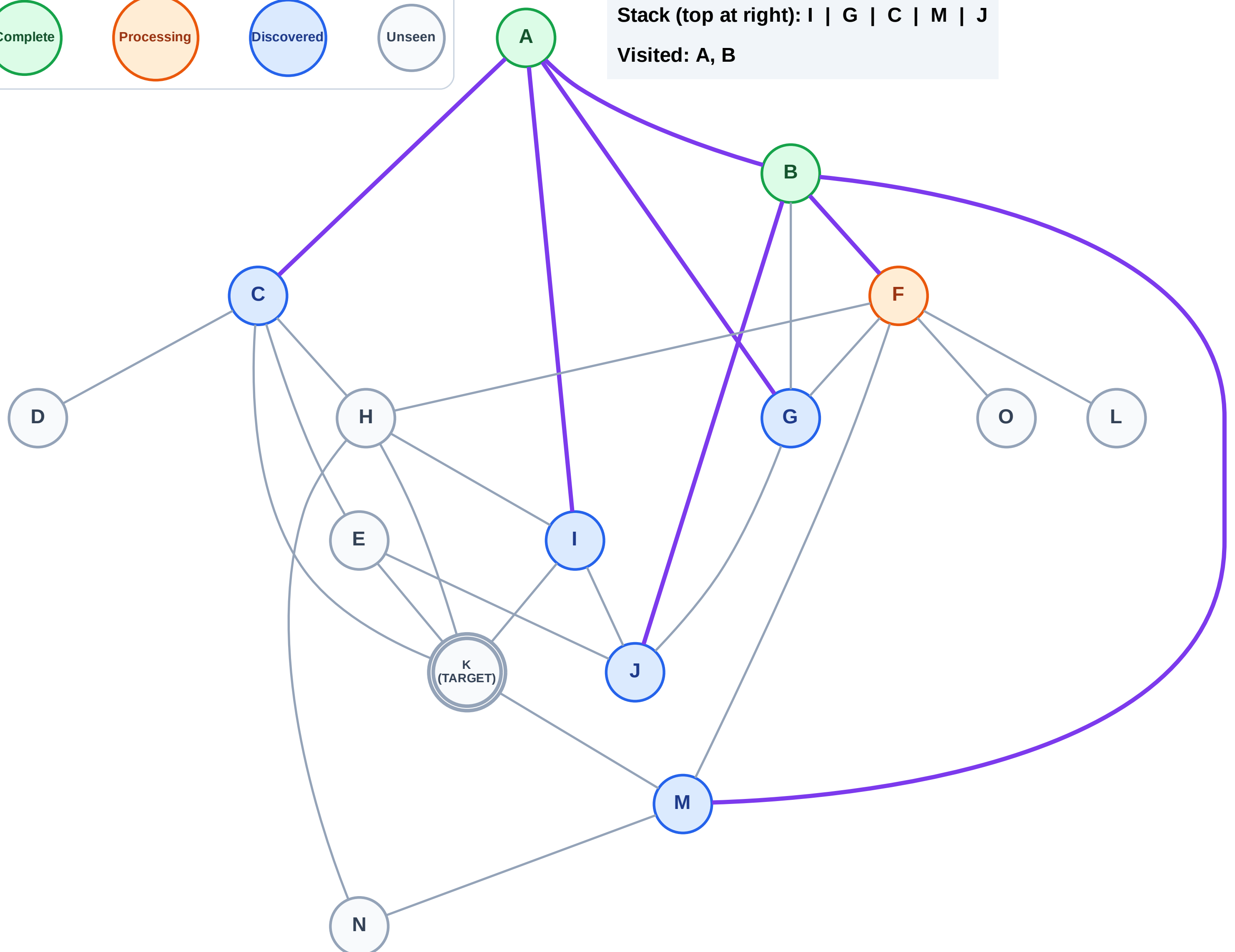
DFS Traversal

Step 6: Process node F



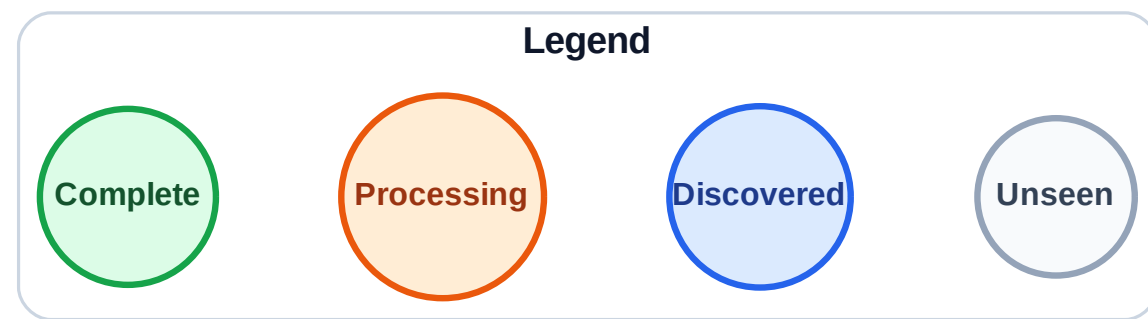
Stack (top at right): I | G | C | M | J

Visited: A, B



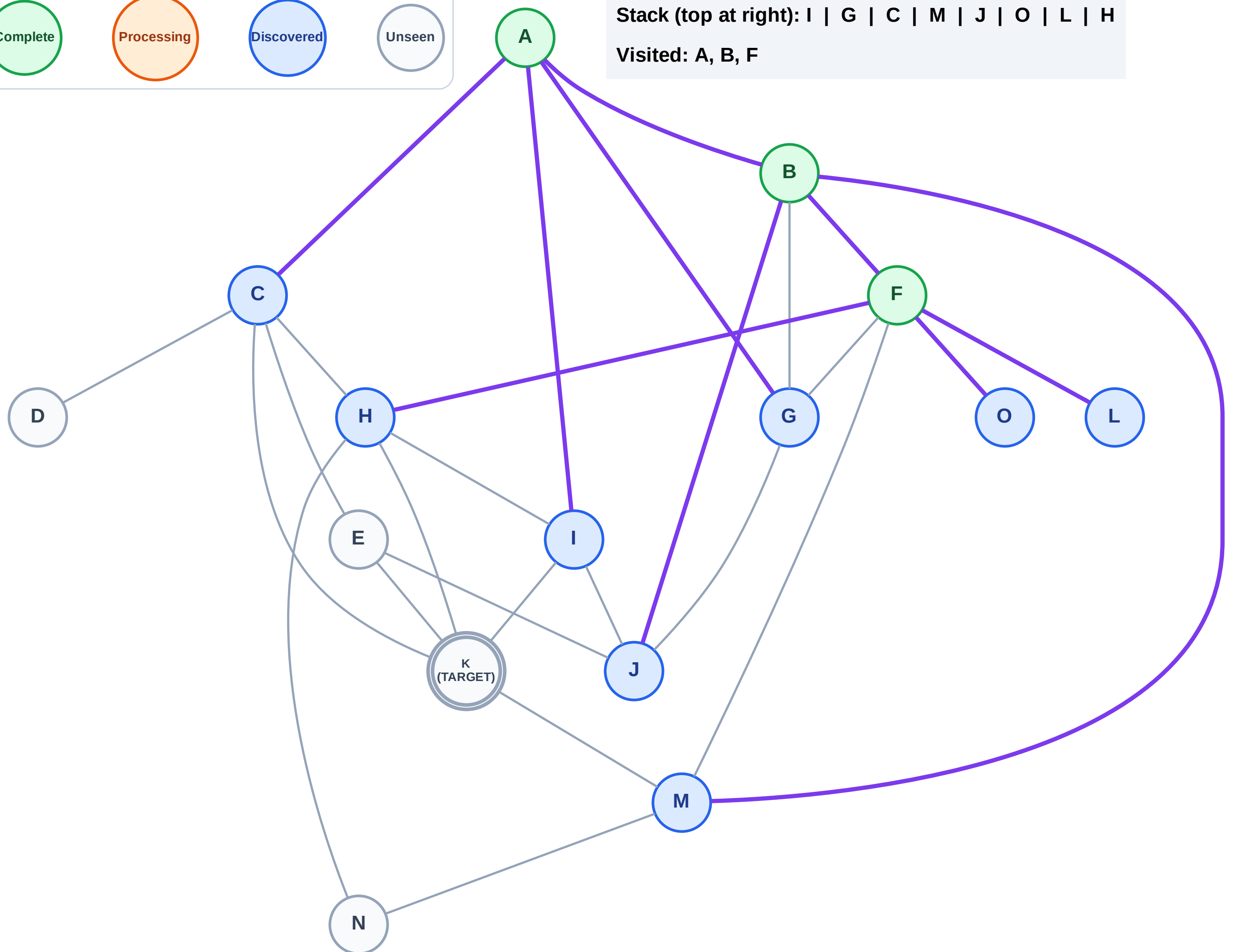
DFS Traversal

Step 7: Discover O, L, H from F



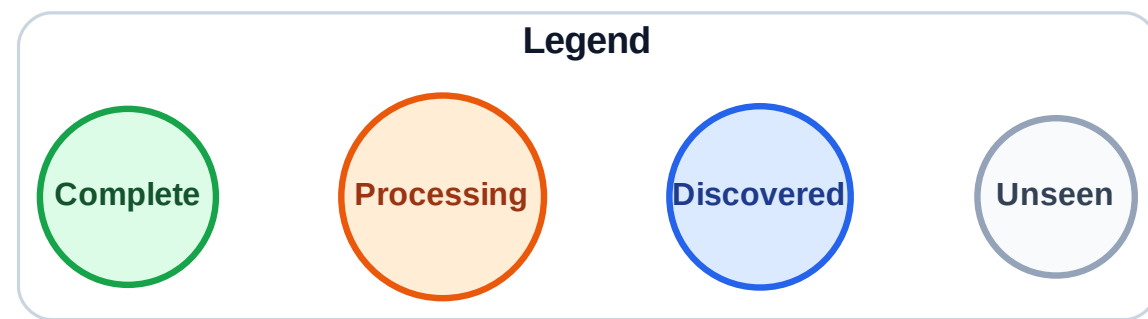
Stack (top at right): I | G | C | M | J | O | L | H

Visited: A, B, F



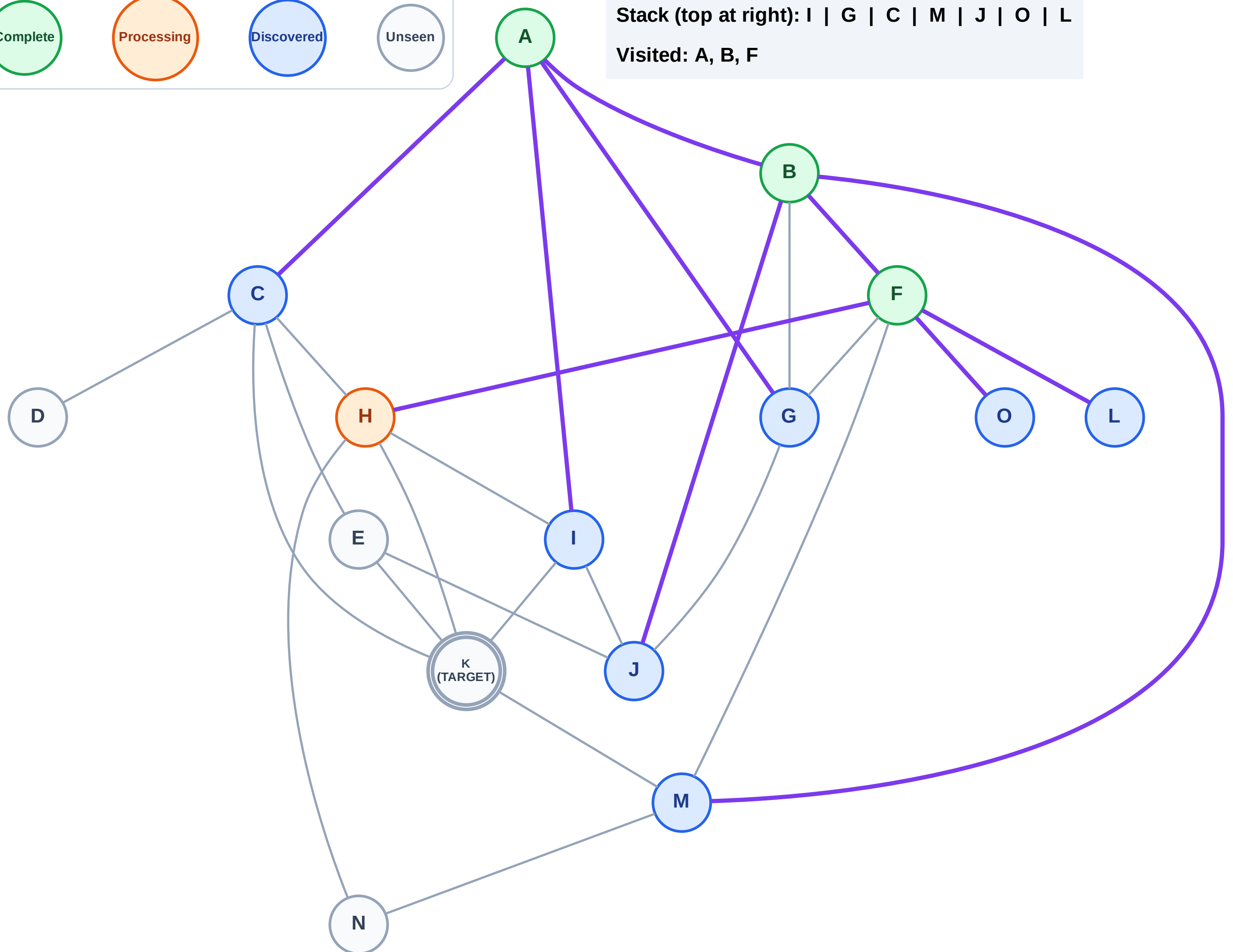
DFS Traversal

Step 8: Process node H



Stack (top at right): I | G | C | M | J | O | L

Visited: A, B, F



DFS Traversal

Step 9: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (4 hops): A → B → F → H → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): I | G | C | M | J | O | L | N

Visited: A, B, F, H, K

